

MOHAMAD ALI SABER

North Kargar. Tehran, Tehran, Iran

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Education

University of Tehran

2019-2023

M.Sc. in Electrical Engineering (Digital Electronic Systems)

Tehran, Iran

- **GPA:** 17.47/20
- **Supervisor:** Dr. Zainolabedin Navabi
- **Thesis:** : Implementation of Near Data Processing Based System for CNN Applications

Shiraz University

2014-2019

B.Sc. in Electrical Engineering

Shiraz, Iran

- **GPA:** 15.19/20

Research Interests

- Computer Architecture
- Computer Arithmetic
- Parallel Processing
- Embedded Systems
- Machine Learning and Deep Learning
- Test and Testability

Publications

- T. Vasei, **M. A. Saber**, A. Nahvy, and Z. Navabi, "An efficient RTL design for a wearable brain-computer interface," IET Computers Digital Techniques (Under Review)
- R. Sadeghi, E. Akbari, and **M. A. Saber**, "On-chip training of crosstalk predictors to fit uncertainties," 2022 IEEE European Test Symposium (ETS), 2022. doi: 10.1109/ETS54262.2022.9810362.
- M. S. Roodsari, **M. A. Saber**, and Z. Navabi, "Diba: N-dimensional bitslice architecture for lstm implementation," 2020 23rd International Symposium on Design and Diagnostics of Electronic Circuits Systems (DDECS), 2020. doi: 10.1109/DDECS50862.2020.9095614 .

Experience

Teaching Experience:

- | | |
|---|-----------------------------------|
| * Teacher Assistance of Digital Logic Design Dr. Navabi. | Spring 2020, Fall 2020 |
| * Teacher Assistance of Digital Logic Design Lab Dr. Navabi. | Fall 2020, Spring 2021 |
| * Teacher Assistance of Microprocessor Dr. Yasrebi. | Spring 2018 |
| * Teacher Assistance of Digital Logic Design Lab Dr. Yasrebi. | Fall 2017, Spring 2018, Fall 2018 |

Work Experience:

- | | |
|---|------------------------|
| * Intern: Researcher in Image and Signal Processing Lab. under supervision of Dr. Yasrebi. | Summer 2018 |
| * junior Developer: junior FPGA(VHDL) Developer in PPAS-Co. | Spring and Summer 2021 |
| * Freelancer: More than 30 academic and industrial projects. | Spring and Summer 2021 |

Awards and Honors

- Summer 2019: Ranked 44th among more than 20,000 participants in the national university entrance exam for master's program in electrical engineering.
- Summer 2014: Rank 1505th regionally among more than 180,000 participants in the national university entrance exam

Relevant Coursework

- | | |
|---|---|
| * <i>Digital System Test and Testability</i> :19.3 out of 20 | * <i>Fault Tolerant System</i> :16.9 out of 20 |
| * <i>Computer Arithmetic</i> :19 out of 20 | * <i>VHDL</i> :17.8 out of 20 |
| * <i>Technologies, Circuits and Memory Systems</i> :17.25 out of 20 | * <i>Computer Architecture</i> :17.27 out of 20 |
| * <i>Microprocessor</i> :19.77 out of 20 | * <i>Interconnects</i> :18.53 out of 20 |

Selected Projects

3D-Printer

2018

- Course: Microprocessors 1 By Dr. N. Yasrebi
- building a 3D printer using AVR and MATLAB.

Design and Implementation of Basic Processor on FPGA

2019

- Undergraduate's Project
- Supervisor: Dr. N. Yasreb
- Implementation a basic computer on FPGA using Xilinx ISE(Verilog) and MATLAB.

Design and Implementation of a Pipeline Processor on FPGA

2019

- Electronics Lab's Project
- Supervisor: Dr. N. Yasreb
- Implementation of a pipeline computer on FPGA using Xilinx ISE(Verilog) and MATLAB.

Design and Implementation of AM/FM Modulator/Demodulator on FPGA

2020

- Supervisor: Dr. M. Farhang
- Implementation of AF and AM modulator and demodulator on FPGA using Xilinx ISE(verilog) and MATLAB.

ISS of several processors such as: AFTAB, PUNEH and SAYAK

2019

- Supervisor: Dr. Z. Navabi
- Developing ISS of AFTAB, PUNEH and SAYAK using C/C++(SystemC).

Certificates

- * **Machine Learning:** (Coursera) Oct. 2023
Verification: <https://www.coursera.org/verify/T383UX3ZHVSU>

Technical Skills

- * **Programming Languages:** C/C++, MATLAB, Python, AVR Assembly, X86 Assembly
- * **Hardware Description Languages:** VHDL, Verilog
- * **System Level Modeling Languages** SystemC, SystemC-AMS
- * **Applications** Xilinx ISE, Xilinx Vivado, Quartus, ModelSim, Codevision, Proteus, HSPICE, Microsoft Office

References

Available on Request